

Customer Churn Prediction

on the IBM Telco Dataset

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Project on Colab

Business Understanding

5-7x

acquiring a new customer costs that much more than keeping one

The problem

Customers cancel their subscription. The company loses the future revenue and the cost of acquiring them.

The goal

Predict in advance who is going to churn, so the retention team can contact them with offers before they leave.

ML problem

Binary classification (Yes/No) plus a probability output to rank customers by risk.

Data Understanding

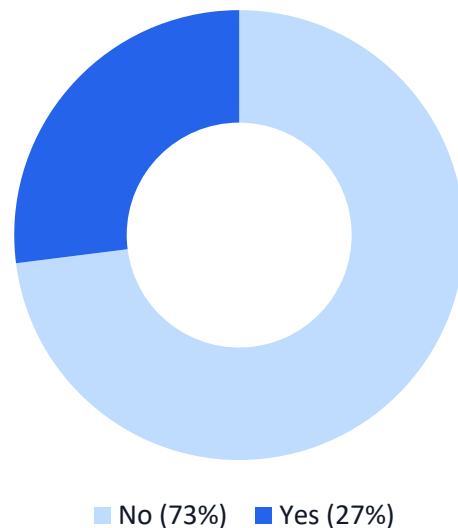
Dataset

- IBM Telco Customer Churn (public)
- 7,043 customers, 21 features
- Target: Churn = Yes / No
- 3 numeric, 16 categorical, 1 ID

Key findings from EDA

- 11 missing TotalCharges (new customers, tenure=0)
- Strong predictors: Contract, tenure, InternetService, PaymentMethod

Class distribution



imbalanced data: accuracy alone is misleading

Data Preparation

Cleaning

- Drop customerID (no predictive value)
- Convert TotalCharges to numeric, drop 11 rows with missing values
- No outliers found (IQR rule, $k=1.5$)

Encoding

- Yes/No -> 0/1
- Contract -> ordinal 0/1/2 (natural order)
- InternetService, PaymentMethod -> one-hot

Train/Test split

- 2/3 train, 1/3 test
- Test set kept aside, only used at the end

Scaling

- StandardScaler for LR and MLP (need scaling)
- Tree models don't need it
- Inside a Pipeline -> no data leakage

Modeling

1

Dummy

baseline (always No)

2

Logistic Regression

linear model

3

Decision Tree

single tree

4

Random Forest

ensemble of trees

5

MLP

neural network

Settings

Hyperparameter tuning: GridSearchCV with 5-fold cross-validation on the training set, F1 (positive class) as scoring

Class imbalance: class_weight='balanced' in LR, Decision Tree and Random Forest (MLP does not support it)

Reason: weighting the minority class makes the model catch more churners (higher recall)

Evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.745	0.508	0.801	0.622	0.842
Random Forest	0.773	0.553	0.700	0.618	0.837
Decision Tree	0.737	0.499	0.749	0.599	0.819
MLP	0.797	0.641	0.510	0.568	0.837
Dummy	0.738	0.000	0.000	0.000	0.500

- Logistic Regression and Random Forest tie for best (F1 ~0.62, ROC-AUC ~0.84). LR is simpler, so it is preferable.
- MLP has the highest accuracy but the worst recall: it misses half the churners. Accuracy is not the right metric here.

Conclusions

Most at-risk customers

- Month-to-month contract
- Short tenure
- Fiber optic internet
- Electronic check payment

Limitations

- Single snapshot, one company
- No time information beyond tenure

